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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

EC2002-1- ANALOG ELECTRONIC CIRCUITS - II

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is the significance of the input offset voltage in an op-amp?**
A) It determines the gain of the amplifier
B) It affects the power dissipation of the op-amp
C) It sets the maximum operating frequency
D) It is the voltage required to make the output exactly zero
2. **A higher slew rate in an op-amp results in:**
A) Faster response time
B) Lower bandwidth
C) Higher input impedance
D) Increased power dissipation
3. **What type of output does a inverting comparator with a reference voltage of 2V give when the input voltage is 3V?**
A) Low (Negative saturation voltage)
B) High (Positive saturation voltage)
C) Zero volts
D) Undefined
4. **If an inverting summing amplifier has three input voltages $V_1 = -2V$, $V_2 = -3V$, $V_3 = -4V$, and the resistors are $R_1 = R_2 = R_3 = 10k\Omega$ and $R_f = 20k\Omega$, what is V_{out} ?**
A) $-18V$
B) $+9V$
C) $-9V$
D) $+18V$
5. **What happens if the clipping voltage in an op-amp clipper is set higher than the input signal's peak voltage?**
A) The signal is completely blocked
B) The output signal is inverted
C) No clipping occurs
D) The circuit stops functioning
6. **What is the primary advantage of a precision rectifier over a conventional diode rectifier?**
A) It requires no external power supply
B) It can rectify very small AC signals
C) It works only with high-frequency signals
D) It does not require any resistors
7. **What should be the order of a filter to have a roll off rate of 100dB/decade in stop band**
A) 3
B) 4
C) 5
D) 6
8. **Identify the order and type of the filter given in Fig mcq.8**

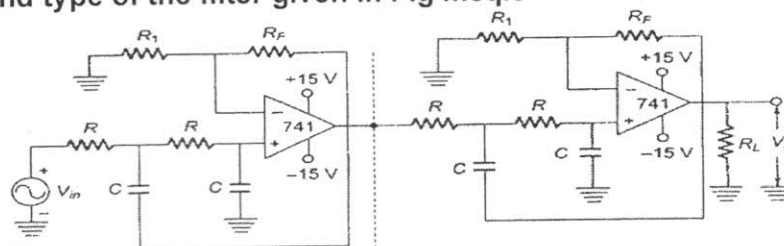


Fig mcq.8

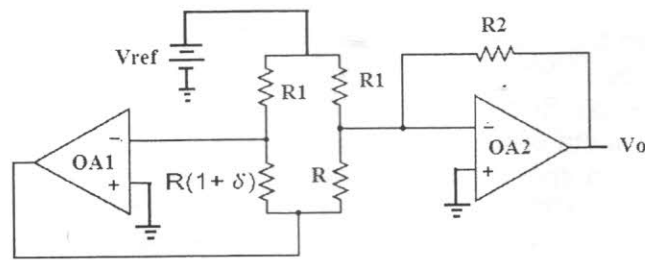
- A) Fourth order high pass filter
- B) Second order band pass filter
- C) Fourth order low pass filter
- D) First order band pass filter

9. What is the advantage of an R-2R ladder DAC compared to a binary-weighted resistor DAC?
- A) Requires fewer resistor values B) Has a faster response time
C) Works without an op-amp D) Uses inductors instead of resistors
10. How many comparators are required for an N-bit Flash ADC?
- A) 2^N B) $2^N - 1$
C) N D) $2N - 1$
11. What happens to the DAC output waveform when a binary counter is used as the digital input?
- A) It produces a stepped increasing waveform B) It generates a sinusoidal signal
C) It remains at a fixed voltage D) It oscillates randomly
12. What is the basic working principle of a linear ramp ADC?
- A) It uses a successive approximation register B) It compares the input voltage with a continuously increasing reference voltage
C) It uses multiple comparators to generate the output in one step D) All of these
13. The control terminal (pin 5) of 555 timer IC is normally connected to ground through a capacitor ($\sim 0.01 \mu\text{F}$). This is to
- A) Protect IC from inadvertent application of high voltage B) Convert the trigger input to sharp pulse by differentiation
C) Prevent false triggering by noise coupled onto the pin D) Suppress any negative triggering pulse
14. In a IC 555 based voltage controlled oscillator, what happens to the output frequency when the control voltage is increased
- A) Increases B) Decreases
C) Remains same D) Becomes zero
15. What is the purpose of the discharge transistor inside the 555 Timer IC?
- A) To charge the external capacitor B) To act as an amplifier
C) To reduce power consumption D) To discharge the external capacitor when required
16. What is the typical voltage level of the output HIGH state in a 555 Timer operating at 9V supply?
- A) 2V B) 9V
C) 4.5V D) 12V
17. How does increasing the overdrive factor affect the switching speed of a transistor
- A) Increases both turn-on and turn-off times B) Reduces both turn-on and turn-off times
C) Reduces turn-on time but increases turn-off time D) Increases turn-on time but reduces turn-off time
18. What happens to the collector-emitter voltage (V_{CE}) during the rise time (t_r)?
- A) It remains constant B) It decreases from a high value towards the saturation voltage ($V_{CE(sat)}$)
C) It increases gradually D) It oscillates between high and low values
19. When the switch in a boost converter is ON, what happens to the inductor?
- A) It discharges its stored energy to the load B) It blocks current flow
C) It steps down the voltage D) It charges and stores energy from the input
20. How does increasing the duty cycle in a boost converter affect the output voltage?
- A) Increases it B) Decreases it
C) Has no effect D) Reduces efficiency

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I****Marks BT* CO* PO***

1. a) Show that the circuit shown in **Fig. Q1a** has output voltage expression

$$V_o = \frac{R_2}{R_1} \delta V_{ref}$$

**Fig. Q1a**

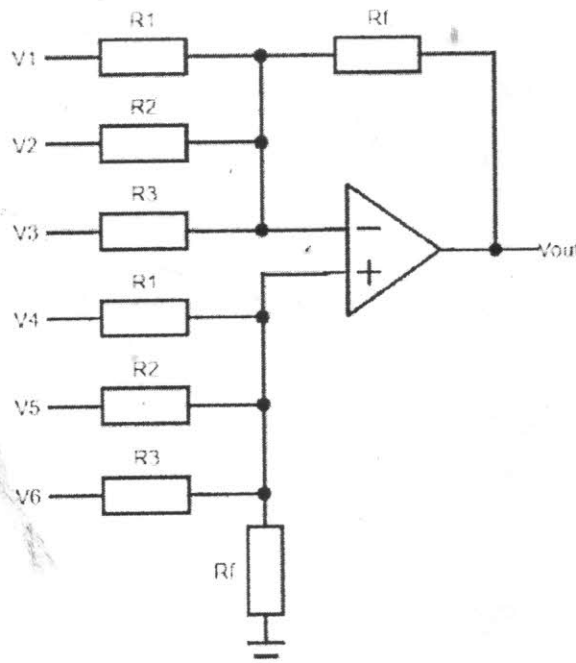
8 L*3 1 1

- b) A band pass filter has $f_L=4\text{kHz}$ and $f_H=12\text{kHz}$ and pass band gain of 9. Calculate the Q value of the filter. Assuming $C=0.05\mu\text{F}$ design the filter such that it has a roll-off rate of 12dB/octave.

8 L3 2 1

2. a) With a neat circuit diagram explain the working of an Instrumentation amplifier. Derive the expression for its gain.
b) For the circuit shown in **Fig. Q2b**. $V_1, V_2, V_3, V_4, V_5, V_6$ are variable DC inputs. Derive the expression for output voltage (V_{out}) if $R_f = 2R_1=4R_2=8R_3$.

8 L2 1 1

**Fig. Q2b**

8 L3 2 1

3. a) With circuit diagram and relevant waveforms explain the working of op-amp based positive clipper.
b) Design a precision FWR with positive gain of 4 and negative gain of -6. Considering $1V_{P.P}$ Sine wave as input, draw the input output waveforms and transfer characteristics.

8 L2 1 1

8 L3 2 1

Unit – II

4. a) Design a R-2R Ladder DAC such that $V_o = 2.25V$ for $B_3B_2B_1B_0 = 0011$ where B_3 is MSB. Also, determine the resolution and full scale output of DAC. 8 L3 3 1
- b) Showing internal diagram draw a IC 555 based Astable multivibrator. Derive the expression for charging and discharging time. 8 L2 4 1
5. a) With a neat diagram explain the working of 3-bit SAR ADC, clearly listing the conversion process. 8 L2 3 1
- b) Using 555 Timer design a Pulse Stretcher circuit that can stretch an input pulse of 10mS to any width of 22mS-110mS. Assume that the capacitor used is $0.1\mu F$. The output voltage variation is 0V-10V. 8 L3 4 1
6. a) With relevant circuit and waveform explain the working of linear ramp ADC. 8 L2 3 1
- b) Explain voltage controlled oscillator using IC555. 8 L2 4 1

Unit – III

7. a) With transient model of BJT and switching waveforms explain how the internal capacitances of the transistor influence the switching characteristics of the transistor. 8 L2 5 1
- b) A simple transistor switch as shown in Fig Q7b is used to connect a 24V dc supply across a relay coil, which has a dc resistance of 200Ω . An input pulse of 0 to 5V amplitude is applied through a series base resistance R_B at the base so as to turn ON the transistor switch. Determine
- I_{CS}
 - Value of R_B required to obtain an ODF of 2
 - Forced β
 - Total power dissipation in the transistor

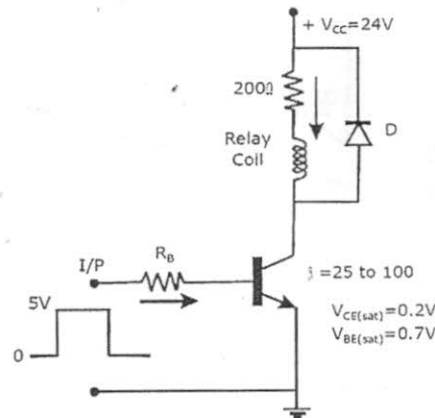


Fig Q7b

8. a) With circuit diagram and waveforms explain the working of the boost regulator. 8 L2 5 1
- b) Explain the Totem pole arrangement for MOSFET gate drive circuit. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
